Yun Hong

COMPUTER SCIENCE | FUDAN UNIVERSITY
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I conduct research at MagicLab, Fudan University, under the guidance of Wenchao Ding. My current research focuses on execution feedback and subtask handoffs for world-action models in long-horizon robotic manipulation.

Education

Fudan University Sep 2023 - Present
B.S. candidate in Computer Science | Shanghai, China

The University of Texas at Austin Aug - Dec 2025
Exchange student in Computer Science | Austin, TX, USA

- GPA: 4.0/4.0 (13 credits). A grades in machine learning, computer vision, operating systems, and probability/random processes.

Research Experience

World-Action Models for Robotic Manipulation Current research
MagicLab, Fudan University | Advisor: Wenchao Ding

- Developing an execution-feedback framework around LingBot-VA on LIBERO-Long; studying how task progress and predicted-future discrepancies can condition the same policy to revise actions.
- Using state replay and controlled interventions to distinguish changes in model outputs from task recovery, and examine the roles of arm/gripper configurations, reachability, and streaming observation history at subtask handoffs.
- Investigating how favorable simulated starting states can be reached through safe, executable actions for subsequent WAM-controlled manipulation.

Risk-Map Learning for Autonomous Driving Oct 2024 - May 2025
MagicLab, Fudan University | Advisor: Wenchao Ding

- Contributed to the Transformer-based risk-prediction framework, model training, debugging, and result verification.
- Fourth author of the RA-L 2026 publication listed below.

Multi-Task Active Perception Fall 2025
RobIn Lab, UT Austin | Advisor: Junhong Xu

- Built multi-task partially observable simulation benchmarks involving unknown object properties and spatial locations.
- Developed VLA baseline training and evaluation workflows to study imitation learning, reinforcement learning, and IL pretraining followed by RL fine-tuning.
- Reviewed manipulation exploration and long-horizon POMDP benchmarks; the project primarily completed benchmark and baseline evaluation workflows.

LLM Reasoning, Reflection, and Self-Correction Sep 2024 - Jun 2025
Fudan NLP | Advisor: Xuanjing Huang

- Constructed self-critic data using Llama3.1-8B-Instruct, with tree-search and reverse-reasoning strategies for diverse correction trajectories.
- Built data synthesis and training workflows; evaluated reasoning and reflection methods on GSM8K.

Publication

Learning a Unified Risk Map for Autonomous Driving in Partially Observable Environments
Jie Jia, Yaofeng Su, Zeyu Bao, **Yun Hong**, Bingzhao Gao, Zhongxue Gan, Wenchao Ding
IEEE Robotics and Automation Letters, 11(4): 4457-4464, 2026.
doi:10.1109/LRA.2026.3666393

Internship Experience

ByteDance Jan - Jun 2026

Test Development Engineer Intern | Douyin R&D, Live Streaming & Media Stream QA

- Developed three AI-assisted tools for routine monitoring, test-case risk checking, and alert summarization.
- Refactored and expanded inspection and test cases. Monitoring accuracy increased from 89% to 100% and coverage from 95% to 98% within the evaluated scope.

Selected Projects

LCDP-Sim: Language-Conditioned Diffusion Policy Started Dec 2025

https://github.com/16yunH/LCDP-Sim

- Built a vision-language-action system with CLIP and a U-Net DDPM/DDIM diffusion policy, including 16-step action chunking.
- Implemented data collection, training, and closed-loop evaluation workflows in ManiSkill2.

MapNavigation Started Dec 2024

https://github.com/16yunH/MapNavigation

- Integrated OpenStreetMap and Gaode Maps API for route planning, location search, hybrid point selection, and road-type speed constraints.

NeuralStyle Started Jun 2025

https://github.com/16yunH/NeuralStyle

- Developed a modular Python style-transfer toolkit with batch processing, configurable experiments, and a browser-based interface.

Skills

Machine learning: Supervised/unsupervised learning, reinforcement learning, imitation learning, model evaluation and generalization.

Deep learning & robotics: PyTorch, Transformers, diffusion models, LLaMA fine-tuning, VLA policies, active perception, POMDPs, and robotic manipulation.

Vision & driving: Multi-view geometry, 3D reconstruction, detection and segmentation, motion planning, trajectory prediction, and risk-field modeling.

Programming & languages: C, C++, Python, LaTeX; Chinese (native), English (fluent).

Honors & Awards

2026: National Second Prize, 3rd “Jinlingguang Cup” China Internet Innovation Competition, University Student Innovation and Entrepreneurship Commercialization Track (team project: Xinghuo Yansheng).

2026: Award recipient, Tencent Rhino-Bird Open-Source Talent Program: OpenCloudOS Security Skill issue (one of three awardees).

2025: Outstanding Work Award, The 5th Meituan Business Analysis Elite Competition.

2024: Shanghai Third Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM).

2024: Finalist, Water, sanitation, and hygiene for the prevention and care of neglected tropical diseases (Geneva, Switzerland).

2023: Finalist, Full-stack AI development engineer skills training by NVIDIA.